

Organic Chemistry Practical 2: Laboratory Techniques & Preparative Organic Synthesis

Reflux Condensation, Simple/Fractional Distillation, Separating Funnel Extraction & Purity Metrics

CANDIDATE NAME	URWAH	CENTRE NUMBER	PK082
COMPONENT	Paper 3 – Advanced Practical Skills (Organic Chemistry Suite)	CANDIDATE NUMBER	0142
TIME ALLOWED	2 Hours	TOTAL MARKS	40 Marks

INSTRUCTIONS TO CANDIDATES:

- Answer all questions in the spaces provided on the question paper.
- Give working for all calculations and record all experimental readings to the required precision:
 - Burette readings to the nearest 0.05 cm³; balance readings to 0.01 g / 0.001 g.
 - Thermometer readings to the nearest 0.5 °C.
- You may use an HB pencil for any rough working, graphs, or apparatus diagrams.
- Non-exact numerical answers should be given to three significant figures unless stated otherwise.
- You will need a scientific calculator and the Qualitative Analysis Notes provided on pages at the back.

LABORATORY INVESTIGATION OVERVIEW:

This laboratory pack contains authentic Cambridge Advanced Practical examinations. Each practical scenario provides complete procedures, chemical specifications (FA codes), hazard warnings, structured data collection tables, graphical plotting grids, and comprehensive multi-step mathematical and observational deductions.

Question 1: Laboratory Preparation and Purification of 1-Bromobutane from Butan-1-ol

Cambridge Syllabus Reference: 9701/35/O/N/22/Q2 · [20 Marks]

Preparative Organic Synthesis Task:

A student prepares 1-bromobutane according to the following reaction:



Experimental Procedure:

1. Place 13.0 g of sodium bromide and 15 cm³ of water into a 100 cm³ round-bottomed flask. Cool the flask in an ice-water bath.
2. Slowly add 10 cm³ of concentrated sulfuric acid with constant swirling.
3. Add 7.40 g (9.15 cm³) of butan-1-ol and 2–3 anti-bumping granules.
4. Fit a water-cooled Liebig condenser vertically in the reflux position. Boil under gentle reflux for 45 minutes.
5. Rearrange the apparatus for distillation and distill until no more oily droplets collect in the receiver.
6. Transfer the distillate into a separating funnel. Separate the organic layer from the aqueous layer.
7. Wash the organic layer with an equal volume of concentrated hydrochloric acid, then with aqueous sodium hydrogencarbonate, and finally with distilled water.
8. Transfer the cloudy organic liquid to a small conical flask, add anhydrous sodium sulfate, and swirl until clear.
9. Decant into a clean pear-shaped flask and redistill, collecting the pure fraction boiling between 101°C and 104°C.

(a) State the purpose of adding anti-bumping granules to the reaction mixture before heating under reflux. [2]

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(b) Explain why the reaction mixture must be heated under reflux rather than in an open flask or beaker. [2]

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(c) State the direction of water flow through the Liebig condenser jacket during reflux and explain why this direction is essential. [2]

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(d) 1-Bromobutane has a density of 1.28 g cm⁻³, whereas water has a density of 1.00 g cm⁻³.

(i) State whether the organic product forms the upper or lower layer in the separating funnel.

(ii) Describe the safety precaution required when shaking the separating funnel with aqueous sodium hydrogencarbonate. [3]

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(e) State the role of anhydrous sodium sulfate in step 8 and state how the student knows by visual inspection that the organic liquid is completely dry. [2]

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(f) In this preparation, 7.40 g of butan-1-ol ($M_r = 74.0$) was used with excess NaBr and H_2SO_4 . The student collected 9.59 g of pure 1-bromobutane ($M_r = 137.0$).

Calculate the theoretical yield of 1-bromobutane in grams and calculate the percentage yield obtained. [5]

(g) Suggest two realistic experimental reasons why the actual percentage yield of 1-bromobutane is significantly below 100%. [4]

Question 2: Distillation vs Reflux in the Oxidation of Ethanol and Determination of Product Purity

Cambridge Syllabus Reference: 9701/33/M/J/21/Q2 · [20 Marks]

Comparative Laboratory Investigation:

The controlled oxidation of ethanol ($\text{CH}_3\text{CH}_2\text{OH}$, b.p. 78°C) by acidified potassium dichromate(VI) can yield either:

- Product A: Ethanal (CH_3CHO , b.p. 21°C)
- Product B: Ethanoic acid (CH_3COOH , b.p. 118°C)

Apparatus Setups:

- Method 1: Ethanol is added dropwise to acidified dichromate in a pear-shaped flask fitted with a still-head, thermometer, and Liebig condenser set for simple distillation. The receiver is immersed in an ice-water bath.
- Method 2: Excess acidified potassium dichromate is heated under reflux with ethanol for 30 minutes, followed by distillation.

Experimental Data:

Boiling points: Ethanal = 21°C , Ethanol = 78°C , Ethanoic acid = 118°C , Water = 100°C .

(a) State which method (Method 1 or Method 2) prepares ethanal as the major product. Explain why the apparatus setup in this method succeeds in preventing ethanal from oxidizing further. [4]

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(b) Explain why the receiving flask in Method 1 must be kept thoroughly cooled in an ice-water bath. [2]

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(c) Write the chemical equation for the oxidation of ethanol to ethanoic acid under reflux in Method 2, using [O] to represent the oxidizing agent. [2]

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(d) State the observation made in the reaction flask during the oxidation of ethanol by acidified potassium dichromate(VI). [2]

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(e) Describe a chemical test on the distillate collected in Method 1 that would confirm the presence of unoxidized ethanol impurity. [3]

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(f) Describe how a student can determine the purity of the synthesized liquid product using a simple boiling-point determination technique. State how the boiling point of an impure liquid compares with that of the pure compound.

[4]

(g) Identify one major hazard associated with handling concentrated sulfuric acid and ethanol during this preparation and specify the appropriate safety measure. [3]

Examiner Mark Scheme & Indicative Observation Guide

Confidential Marking Points & Error Tolerances for Organic Chemistry Practical 2: Laboratory Techniques & Preparative Organic Synthesis

Question 1: Laboratory Preparation and Purification of 1-Bromobutane from Butan-1-ol [20 Marks]

Part	Marking Points & Expected Observations	Marks
(a)	To provide nucleation sites for smooth bubble formation [1], preventing violent boiling, uneven surging, or 'bumping' of the liquid out of the flask [1].	[2]
(b)	To prevent the loss of volatile organic reactants (butan-1-ol) and products (1-bromobutane) by evaporation [1], while allowing the reaction to proceed continuously at its boiling temperature [1].	[2]
(c)	Water enters at the bottom (lower inlet) and leaves at the top (upper outlet) [1]; this ensures the condenser jacket fills completely with cold water without trapping air bubbles or air pockets, ensuring efficient cooling [1].	[2]
(d)	(i) Lower layer (since its density of 1.28 g cm^{-3} is greater than water) [1]. (ii) Periodically invert the separating funnel and open the stopcock / tap [1] to release pressure buildup caused by the evolution of carbon dioxide gas (CO_2) from neutralizing acid residues [1].	[3]
(e)	Role: Acts as a chemical drying agent to absorb and remove trace dissolved water [1]; Visual check: The liquid turns from turbid / cloudy / milky to completely clear / transparent, and the solid drying agent moves freely without sticking / clumping to the glass [1].	[2]
(f)	Moles of butan-1-ol = $7.40 / 74.0 = 0.100 \text{ mol}$ [1]. Mole ratio of butan-1-ol to 1-bromobutane is 1:1, so theoretical moles = 0.100 mol [1]. Theoretical yield = $0.100 \times 137.0 = 13.70 \text{ g}$ [1]. Actual yield = 9.59 g [1]. Percentage yield = $(9.59 / 13.70) \times 100 = 70.0\%$ (allow $70.0\% - 70.1\%$) [1].	[5]
(g)	1. Incomplete reaction / reversible equilibrium or side reactions occurring (e.g. elimination to form but-1-ene gas or condensation to form di-n-butyl ether) [2]. 2. Mechanical losses during transfers between glassware (flask to condenser, separating funnel stopcock, conical flask drying, redistillation residue left in boiling flask) [2].	[4]

Question 2: Distillation vs Reflux in the Oxidation of Ethanol and Determination of Product Purity [20 Marks]

Part	Marking Points & Expected Observations	Marks
(a)	Method 1 prepares ethanal [1]. Explanation: Ethanal has a very low boiling point (21°C) compared to ethanol (78°C) and ethanoic acid (118°C) [1]. In Method 1 (distillation), as soon as ethanal forms, it vaporizes immediately and distills over out of the reaction flask [1], removing it from contact with the oxidizing agent before it can be further oxidized to ethanoic acid [1].	[4]
(b)	Ethanal has a boiling point of only 21°C (close to room temperature) and high volatility [1]; ice-cooling condenses the vapor and prevents ethanal escaping as flammable vapor into the laboratory [1].	[2]
(c)	$\text{CH}_3\text{CH}_2\text{OH} + 2[\text{O}] \rightarrow \text{CH}_3\text{COOH} + \text{H}_2\text{O}$ [2] (1 mark for correct formulas, 1 mark for balancing $2[\text{O}]$ and H_2O).	[2]
(d)	The solution turns from orange ($\text{Cr}_2\text{O}_7^{2-}$) to dark green (Cr^{3+}) [2] (1 mark for orange, 1 mark for green).	[2]
(e)	Add a small dry piece of sodium metal: effervescence / bubbles of hydrogen gas produced confirms ethanol [2] (or add PCl_5 : steamy acidic fumes of HCl that turn damp blue litmus red confirms $-\text{OH}$ group [2]).	[3]
(f)	Technique: Set up a distillation apparatus with a sensitive thermometer placed with its bulb level with the side-arm condenser entrance [1], or use a micro-boiling tube method (Siwoloboff's method with inverted capillary) [1]. Pure compound: Distills at a sharp, constant boiling point identical to literature value (e.g. exactly 118°C for ethanoic acid) [1]. Impure compound: Boils over a broad temperature range and the observed boiling point is elevated or shifted compared to the pure substance [1].	[4]

(g)	Hazard: Concentrated sulfuric acid is highly corrosive and causes severe skin burns; reaction with alcohol is violently exothermic; ethanol is highly flammable [2]. Safety measure: Add acid slowly with continuous cooling in ice and swirling; wear chemical-resistant nitrile gloves and goggles; keep ethanol away from open Bunsen flames by using an electric heating mantle or water bath [1].	[3]
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